

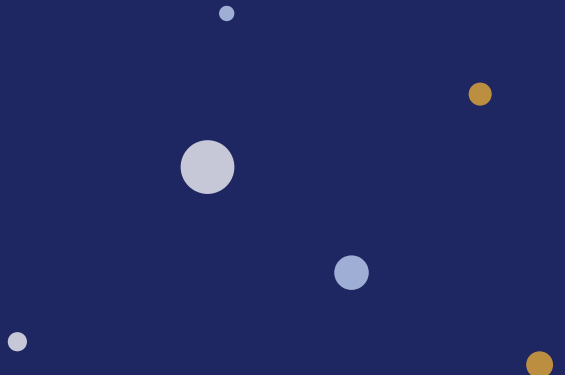


10-DAY STUDY PLAN

Pandas for SQL Professionals

Porting 25+ years of SQL fluency (Oracle, SQL Server, DB2, MySQL, PostgreSQL, MongoDB) to Python, Pandas, and Snowflake.

Companion deck: "Pandas — Complete Study Guide & Cheatsheet"



Why Pandas, Why Now

- Pandas is Python's core library for structured data manipulation, cleaning, and analysis — built on NumPy, and the on-ramp to matplotlib, seaborn, and scikit-learn.
- Data has moved onto cloud platforms — Snowflake, Databricks, AWS/Azure/GCP — and modern analytics work now expects Python alongside SQL.
- You already think in tables, keys, and set operations. That instinct transfers almost directly onto the DataFrame model.
- **Goal of this plan: reuse your SQL mental model as the fastest path to real Pandas fluency in 10 focused days.**

The SQL-to-Pandas Mental Model

SQL Concept	Pandas Equivalent
Table	DataFrame
Column	Series
Row / primary key	Index (row label)
WHERE	Boolean mask, <code>.query()</code>
GROUP BY / HAVING	<code>.groupby().agg()</code> , then filter
JOIN (inner/left/right/outer)	<code>.merge(how=...)</code>
UNION / UNION ALL	<code>pd.concat()</code>
ORDER BY	<code>.sort_values()</code>
Window functions (OVER/PARTITION BY)	<code>.groupby().transform()</code> , <code>.rolling()</code> , <code>.rank()</code>
PIVOT / UNPIVOT	<code>.pivot_table()</code> / <code>.melt()</code>
CASE WHEN	<code>np.where()</code> / <code>np.select()</code>

How to Use This Plan

- Budget ~1.5–2.5 hours per day: watch or read the linked resource first, then complete the hands-on exercise in your own Jupyter or VS Code notebook.
- Bring your own data from Day 2 onward if you can — nothing cements Pandas like running it against domain tables you already know.
- Keep a personal "SQL → Pandas" translation notebook: for every SQL query you already know, write the Pandas equivalent beside it.
- Days build on each other — don't skip Day 1–2. The index/dtype model is the one place Pandas genuinely differs from SQL.

10-Day Overview

Day	Focus	SQL Analogy Anchor
1	Environment setup, Series & DataFrame basics	Table & column model
2	Data I/O: CSV / Excel / JSON / SQL / Snowflake	SELECT * FROM table
3	Selection, indexing, filtering	WHERE
4	Sorting & derived columns	ORDER BY, computed SELECT
5	GroupBy & aggregation	GROUP BY / HAVING
6	Merging, joining, concatenating	JOIN, UNION
7	Data cleaning	ETL data-quality scrubbing
8	Reshaping data	PIVOT / UNPIVOT
9	Analysis: stats, correlation, windows	Window functions, aggregates
10	Visualization + capstone project	Reporting layer



DAY 1

Environment & Core Data Structures

SQL anchor: *the table & column model*

LEARNING OBJECTIVES

- Install Python, Pandas, Jupyter/VS Code, and snowflake-connector-python[pandas]
- Understand Series (one typed column + index) and DataFrame (a table of Series)
- Inspect data: `.shape`, `.dtypes`, `.info()`, `.describe()`, `.head()`/`.tail()`

RESOURCES

- [Official: 10 Minutes to pandas](#)
- [Book: Python for Data Analysis, 3E, Ch. 5](#)

HANDS-ON EXERCISE

Build a DataFrame from a dict representing a table you know well (e.g. claims or members). Run `.info()` and `.describe()` and compare to what DESCRIBE TABLE would tell you.

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DAY 2

Data I/O — Getting Data In and Out

SQL anchor: `SELECT * FROM table`

LEARNING OBJECTIVES

- Read/write CSV, Excel, JSON with `read_csv/read_excel/read_json` and key options (`dtype`, `parse_dates`, `chunksize`)
- Pull data straight from Snowflake with `pandas.read_sql` or `fetch_pandas_all()`
- Write a DataFrame back to Snowflake with `write_pandas()`

RESOURCES

- [Snowflake: Using pandas DataFrames with the Python Connector](#)
- [pandas I/O guide](#)

HANDS-ON EXERCISE

Connect to a Snowflake sandbox schema and pull one familiar table into a DataFrame with `read_sql`. No access handy? Export to CSV and load that instead. Confirm dtypes match expectations.



DAY 3

Selection, Indexing & Filtering

SQL anchor: *WHERE*

LEARNING OBJECTIVES

- Column selection: `df['col']` vs `df[['a','b']]`
- Row selection: `.loc` (label-based) vs `.iloc` (position-based)
- Boolean masks and `.query()` as WHERE-clause equivalents; combine with `&` | `~`

RESOURCES

- [Official Comparison with SQL \(bookmark this\)](#)
- [Video: Keith Galli, Complete Pandas Tutorial](#)

HANDS-ON EXERCISE

Take 5 SQL WHERE clauses you've written recently (multi-condition, IN, BETWEEN, LIKE) and translate each into Pandas — once with `.loc` + masks, once with `.query()`.

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DAY 4

Sorting & Derived Columns

SQL anchor: *ORDER BY*, *computed SELECT*

LEARNING OBJECTIVES

- `.sort_values(by=[...], ascending=[...])` and `.sort_index()`
- Computed columns via `.assign()`, vectorized arithmetic, and `.apply()` (a last resort, not a first choice)
- CASE WHEN equivalents: `np.where()` for binary, `np.select()` for multi-branch

RESOURCES

- [Data School \(Kevin Markham\) — Easier data analysis with pandas](#)

HANDS-ON EXERCISE

Recreate a ``SELECT *, CASE WHEN ... END AS flag ORDER BY col DESC`` report — build the flag with `np.where/np.select` and sort with `sort_values`.

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DAY 5

GroupBy & Aggregation

SQL anchor: *GROUP BY / HAVING*

LEARNING OBJECTIVES

- The split-apply-combine model behind `.groupby()`
- `.agg()` with multiple functions per column; named aggregation with `pd.NamedAgg`
- HAVING equivalent: filter the grouped result, or `.groupby().filter()`

RESOURCES

- [Official groupby guide](#)
- [Video: freeCodeCamp, Data Analysis with Python \(full course\)](#)

HANDS-ON EXERCISE

Rebuild a ``GROUP BY key1, key2 HAVING COUNT(*) > n`` report from a domain table, using `groupby` + `agg` + a post-filter.



DAY 6

Merging, Joining & Concatenating

SQL anchor: *JOIN, UNION*

LEARNING OBJECTIVES

- `.merge(how='inner' | 'left' | 'right' | 'outer', on=/left_on=/right_on=)`
- `validate=` and `indicator=True` to catch cardinality surprises before they corrupt a report
- `pd.concat()` for UNION/UNION ALL-style row stacking; `.join()` for index-based joins

RESOURCES

- [Official Comparison with SQL — Join section](#)
- [GitHub: Keith Galli, complete-pandas-tutorial](#)

HANDS-ON EXERCISE

Pick 2–3 interconnected domain tables you join in SQL routinely. Reproduce the join chain with `.merge()`, using `indicator=True` to confirm row counts match exactly.



DAY 7

Data Cleaning

SQL anchor: *ETL scrubbing / data-quality rules*

LEARNING OBJECTIVES

- Missing data: `.isna()/.notna()`, `.fillna()` strategies, `.dropna()`, `.interpolate()`
- Duplicates: `.duplicated()`, `.drop_duplicates()`
- Type conversions: `.astype()`, `to_datetime()`, `to_numeric()`; category dtype; `.str` cleaning

RESOURCES

- [Official: Working with missing data guide](#)
- [Book: Python for Data Analysis, 3E, Ch. 7](#)

HANDS-ON EXERCISE

Take a genuinely messy extract (blank values, inconsistent casing, duplicate rows) and produce a clean DataFrame, documenting each cleaning decision.



DAY 8

Reshaping Data

SQL anchor: *PIVOT / UNPIVOT*

LEARNING OBJECTIVES

- `.pivot_table()` — PIVOT with an aggregation function built in
- `.melt()` — UNPIVOT, wide-to-long
- `.stack()/unstack()` for MultiIndex reshaping; `pd.crosstab()` for quick frequency tables

RESOURCES

- [Official reshaping guide](#)

HANDS-ON EXERCISE

Reproduce a wide PIVOT-style report with `pivot_table`, then `melt()` it back to long form and confirm you recover your original rows.



DAY 9

Analysis — Stats, Correlation & Window Functions

SQL anchor: *OVER (PARTITION BY ... ORDER BY ...)*

LEARNING OBJECTIVES

- Descriptive stats: `.describe()`, `.mean()/.std()`, `.value_counts()`
- `.corr()` for correlation matrices between numeric columns
- Window-function equivalents: `.rolling()`, `.expanding()`, `.shift()`, `.rank()`, `.cumsum()`

RESOURCES

- [Official window operations guide](#)
- [Video: Corey Schafer, Pandas Tutorials playlist](#)

HANDS-ON EXERCISE

Recreate a running-total or ranking report you've built with `SUM(...) OVER (PARTITION BY ...)` using `groupby` + `rolling/cumsum/rank`.

Visualization + Capstone Project

SQL anchor: *the reporting layer*

LEARNING OBJECTIVES

- Quick plotting straight from Pandas: `df.plot(kind=...)`
- Matplotlib fundamentals for full control over figures and axes
- Seaborn for statistical charts: distributions, correlation heatmaps, category comparisons

RESOURCES

- [Video: freeCodeCamp full Data Analysis course](#)
- [Official visualization guide](#)

HANDS-ON EXERCISE

See the dedicated capstone project on the next slide — this is the day it all comes together.

Capstone Project (2–3 hrs)

STEPS

- Pull 2–3 related tables from Snowflake (or CSV) into DataFrames
- Clean each: missing data, dtypes, duplicates
- Join them into one analysis-ready DataFrame
- Group/aggregate to answer a real question from your domain (claims, care management, etc.)
- Add a derived column and one window-style calculation (e.g. month-over-month change)
- Visualize the result with at least two chart types
- Write the result back to Snowflake or export to Excel for sharing

RESOURCES FOR TODAY

- freeCodeCamp — Data Analysis with Python (NumPy, Pandas, Matplotlib, Seaborn)
- Official Pandas visualization guide
- Snowflake official: `write_pandas()` and `pandas-on-Snowflake` docs
- **This capstone is the dress rehearsal for real work — treat the output as something you'd actually hand to a colleague.**

KEEP THESE HANDY

Resource Summary — Quick Links

pandas official documentation pandas.pydata.org/docs

Comparison with SQL (official)

pandas.pydata.org/docs/getting_started/comparison/comparison_with_sql.html

Official pandas cheat sheet (PDF) pandas.pydata.org/Pandas_Cheat_Sheet.pdf

Free book — Python for Data Analysis, 3E wesmckinney.com/book

Keith Galli — Complete Pandas Tutorial github.com/KeithGalli/complete-pandas-tutorial · youtu.be/2uvysYbKdjM

Corey Schafer — Pandas Tutorials playlist youtube.com/playlist?list=PL-osiE80TeTsWmV9i9c58mdDCSskIFdDS

Data School — Easier data analysis with pandas dataschool.io/easier-data-analysis-with-pandas

freeCodeCamp — Data Analysis with Python (full course)
freecodecamp.org/news/how-to-analyze-data-with-python-pandas

Snowflake + pandas (official) docs.snowflake.com/en/developer-guide/python-connector/python-connector-pandas

pandas on Snowflake — Snowpark pandas API
docs.snowflake.com/en/developer-guide/snowpark/python/pandas-on-snowflake

Where to Go Next

- Keep the companion "Pandas — Complete Study Guide" deck as your reference during and after the 10 days.
- Try the Snowpark pandas API (Modin-based) to run pandas syntax that executes inside Snowflake at scale, once local pandas feels natural.
- Add scikit-learn for modeling once your prep pipeline is solid — pandas stays your prep layer.
- Revisit Day 6 (joins) and Day 9 (window functions) after two weeks of real use — that's where SQL habits most often cause pandas bugs.